

Machine learning for vision

Winter 2016

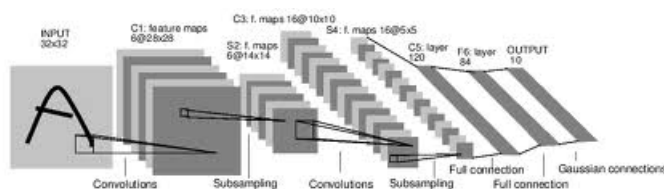
Roland Memisevic

Lecture 3, February 2, 2016

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Machine learning for vision

Convolutional neural networks (CNN)

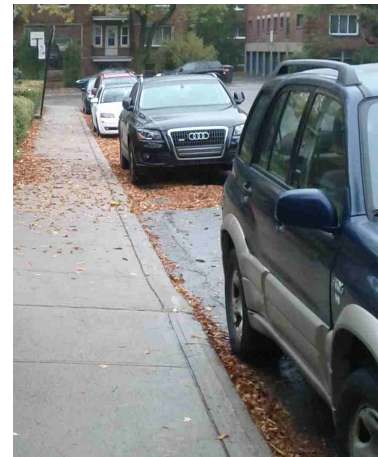


- ▶ Two main ideas:
 1. Use only **local** features.
 2. Apply **the same** features at many different positions.
- ▶ Add subsampling (max- or average-pooling) layers, so that higher layers can see (gradually) larger regions of the image.
- ▶ Standard reference: LeCun et al. 1998

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Translation invariance and locality

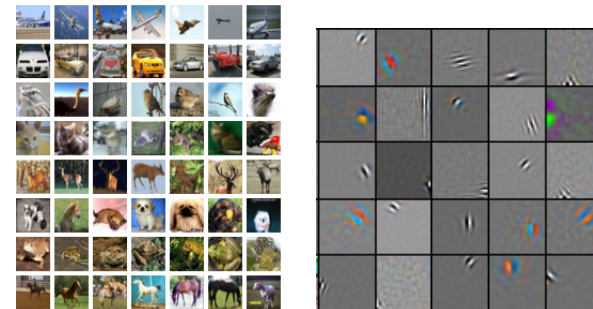


- ▶ Most structure in natural images is **local** → Most low-level operations may be based on patches.
- ▶ Most structure in natural images is **position-invariant** → We may perform the same set operations everywhere across the image.

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Features *want* to be local

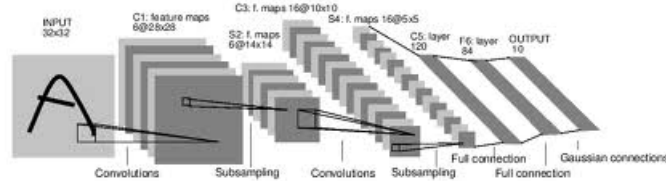


- ▶ (right) Features learned by a fully connected network on RGB images of size 32×32 pixels.
- ▶ A reasonable number of features for these images might be $32 \times 32 \times 3 = 3072$ (a complete basis)
- ▶ Better would be overcomplete.
- ▶ → you should use ≥ 9 mio parameters in the first layer

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Convolutional neural networks (CNN)



- ▶ A convolution layer operates on 3-D tensors:
 $\text{num vertical pixels} \times \text{num horizontal pixels} \times \text{num channels}$.
- ▶ The number of channels in the input can be one (gray value image), 3 (RGB image), or anything else. In higher layers it is usually more.

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Neural network activations are sparse

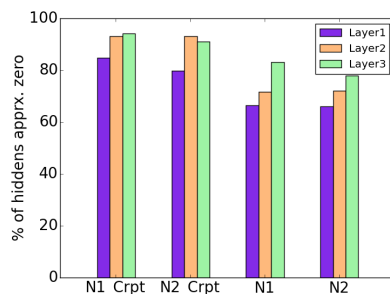


figure by Kishore Konda

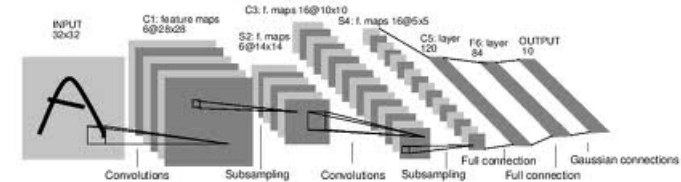
Sparsity levels in two networks trained on CIFAR-10. N1=(1000-2000-3000), N2=(2000-2000-2000 units). (N1_Crpt, N2_Crpt trained with dropout).

- ▶ Sparsity can lead to “update scarcity”.
- ▶ More neurons are good but naturally lead to more sparsity.
- ▶ This is likely why weight sharing works so well in conv-nets.

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Convolutional neural networks (CNN)



- ▶ The number of parameters in a convolutional layer is:
 $\text{num vertical filtersize} \times \text{num horizontal filtersize} \times \text{num input channels} \times \text{num output channels}$
- ▶ Without weight sharing, this could be, say,
 $10 \times 10 \times 3 \times 3072 \approx 1 \text{ mio}$ (if complete)
- ▶ With weight sharing this could be
 $10 \times 10 \times 3 \times 100 \approx 30\text{k}$ (already overcomplete)
- ▶ The number of *hidden units* in this case:
 $\text{num vertical pixels} \times \text{num horizontal pixels} \times \text{num channels}$ (eg. $32 \times 32 \times 100 = 102400$)

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Convolutional neural networks (CNN)

- ▶ Conv-nets maximize *the ratio*

$$\frac{\# \text{neurons}}{\# \text{parameters}}$$
- ▶ Their good performance is likely due to
 - ▶ many neurons being better at untangling the variability in the data, and
 - ▶ weight sharing compensating for the resulting update scarcity
- ▶ Caveat: During learning, updates need to be distributed across parameters, which requires long-range communication, so it doesn't map nicely to very simple hardware and is not plausible biologically.
- ▶ (During inference, filters are usually distributed over image locations, too, although in principle one could use many copies of the same filter instead).

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ImageNet challenge 2012

ImageNet Large Scale Visual Recognition Challenge 2012 (ILSVRC2012)
Held in conjunction with PASCAL Visual Object Classes Challenge 2012 (VOC2012)

Back to Main page

All results

- Task 1 (classification)
- Task 2 (localization)
- Task 3 (fine-grained classification)
- Team information and abstracts

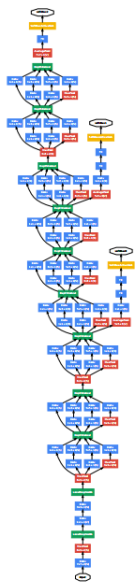
Task 1

Team name	Filename	Error (9 guesses)	Description
SuperVision	test_preds-141-146-2009-131-137-145-146-2011-145f	0.15315	Using extra training data from ImageNet Fall 2011 release
SuperVision	test_preds-131-137-145-135-143f.txt	0.16422	Using only supplied training data
ISI	pred_FVv_wLACs_weighted.txt	0.26172	Weighted sum of scores from each classifier with SIFT+PV, LBP+PV, GIST+PV, and CSIFT+PV, respectively.
ISI	pred_FVv_weighted.txt	0.26602	Weighted sum of scores from classifiers using each FV.
ISI	pred_FVv_summed.txt	0.26646	Naive sum of scores from classifiers using each FV.
ISI	pred_FVv_wLACs_weighted.txt	0.26653	Naive sum of scores from each classifier with SIFT+PV, LBP+PV, GIST+PV, and CSIFT+PV, respectively.

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GoogLeNet (Szegedy et al. 2014)



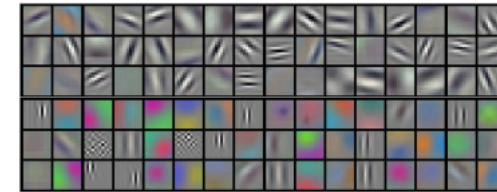
- ▶ Won ImageNet 2014 with **6.66%** top-5 error rate
- ▶ Key insights: Scaling up, unconventional architecture (eg. cross-channel pooling), intermediate targets
- ▶ A variation of this network based on *Batch Normalization* (Ioffe, Szegedy, 2015) achieves **4.8%** top-5 error rate, surpassing the accuracy of human labelers

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ImageNet challenge 2012

some first-layer features



some results

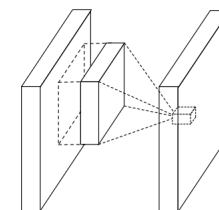


Krizhevsky, Sutskever, Hinton; 2012

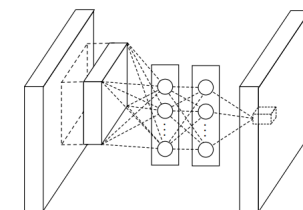
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Network in Network



(a) Linear convolution layer



(b) Mlpconv layer

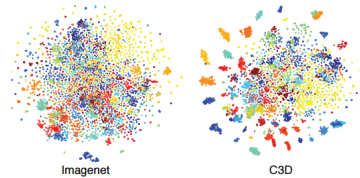
- ▶ Lin, Chen, Yan 2013

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Video convnets: C3D

Dataset Task	Sport1M action recognition	UCF101 action recognition	ASLAN action similarity labeling	YUPENN scene classification	UMD scene classification	Object object recognition
Method	[19]	[39][26]	[31]	[10]	[10]	[32]
Result	80.2	75.8 (89.1)	68.7	96.2	77.7	12.0
C3D	85.2	85.2 (90.4)	78.3	98.1	87.7	22.3
Δ	5.0	9.4 (1.3)	9.6	1.9	10.0	10.3



(Tran, Bourdev, Fergus, Torresani, Paluri; 2015)

High-level features

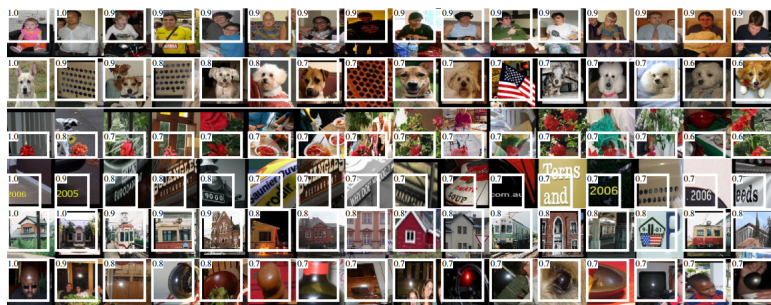
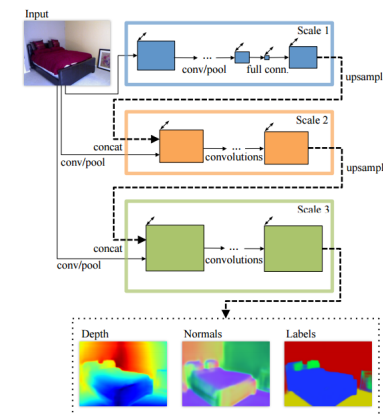


Figure 4: Top regions for six pool units. Receptive fields and activation values are drawn in white. Some units are aligned to concepts, such as people (row 1) or text (4). Other units capture texture and material properties, such as dot arrays (2) and specular reflections (6).

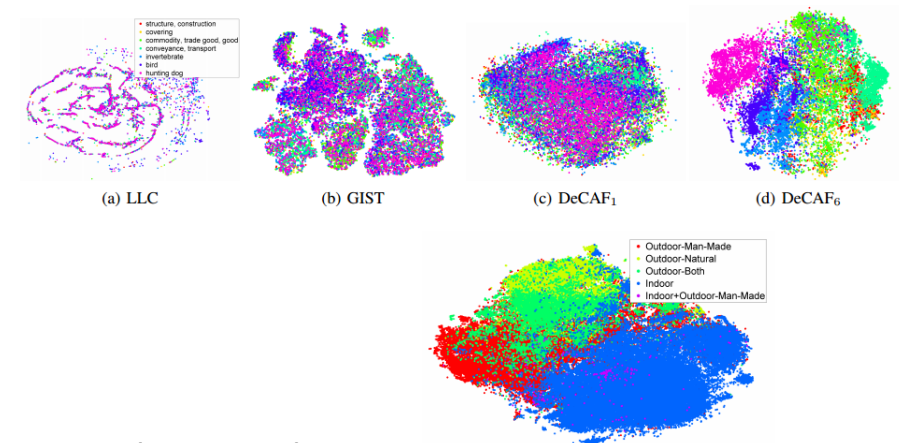
Girshick et al., 2014

Predicting depth, surface normals, semantic labels



Eigen, Fergus (2014)

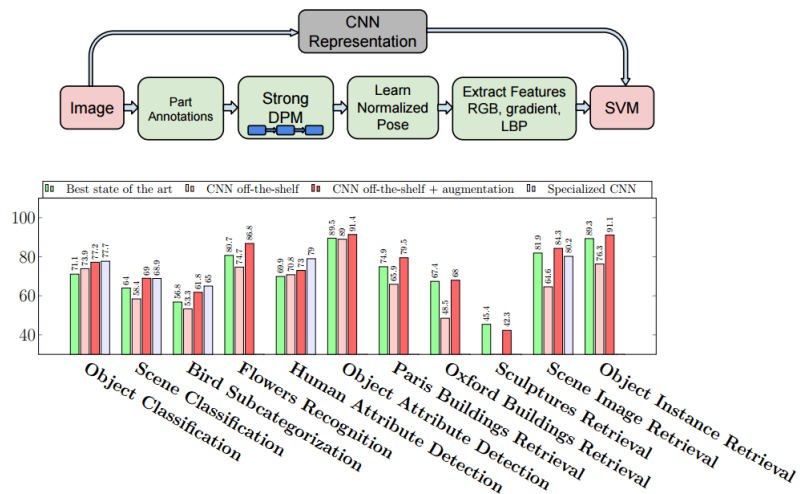
Conv-nets learn good *generic* features



non-imagenet classes:

(Donahue et al, 2013)

Conv-nets learn good *generic* features



(Razavian, Azizpour, Sullivan, Carlsson; 2014)

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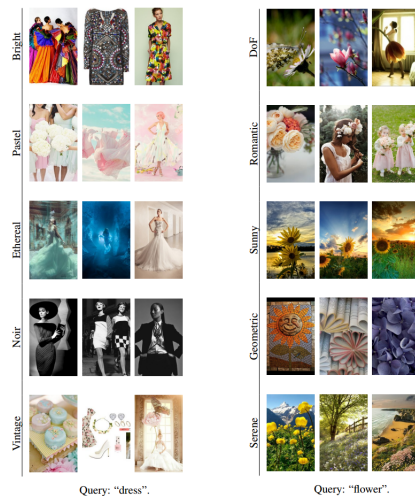
Karayev et al 2014: Recognizing Image Style



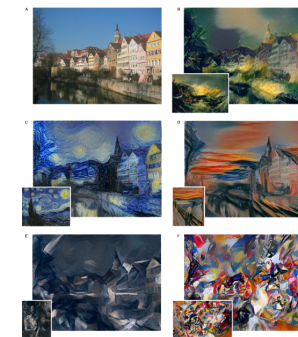
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Karayev et al 2014: Recognizing Image Style



Rendering images using a pre-trained convnet



Gatys, Ecker, Bethge (2015)

- Idea: optimize pixels so as to (i) match the hidden layer activations of the content image (ii) match a non-linear function of the hidden layer activations of the style image

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Off-the-shelf networks

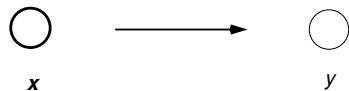
- ▶ It has become hard to compete with features from a pre-trained network (such as GoogLeNet) on almost any recognition task.
- ▶ As a result, designing and training your own conv-net (similarly language-, translation-, speech-model, etc.) is becoming harder and less desirable.
- ▶ Pre-trained, off-the-shelf networks are therefore likely to become much more important in the future.
- ▶ The upside: this leads to a separation-of-concerns that gives AI researchers a huge advantage over evolution.

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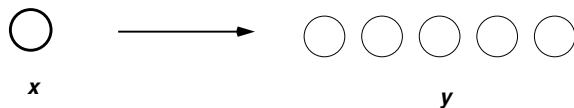
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Structured prediction

Prediction:

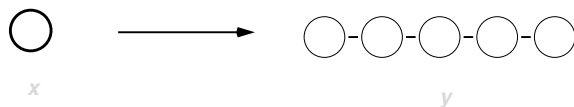


Structured prediction:



Problem: combinatorial explosion

Solution: Impose tractable dependency structure



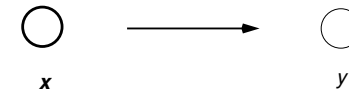
- ▶ Applications: Scene labeling, text, speech, ...

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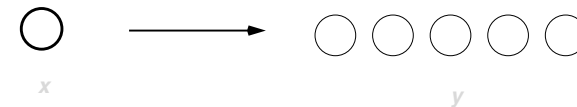
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Structured prediction

Prediction:

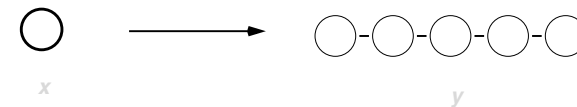


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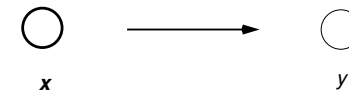
- ▶ Applications: Scene labeling, text, speech, ...

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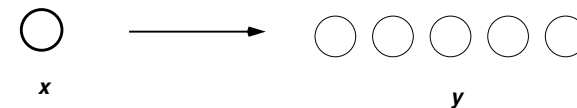
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Structured prediction

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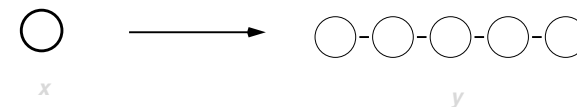


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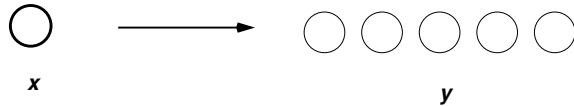
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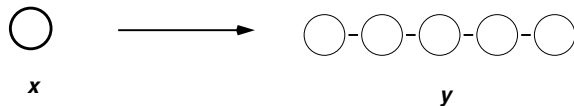


Structured prediction:



Problem: combinatorial explosion

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- Applications: Scene labeling, text, speech, ...

Linear structured prediction

- Main idea: Define scores over cliques in a graph
- Example: sequence models may define a score

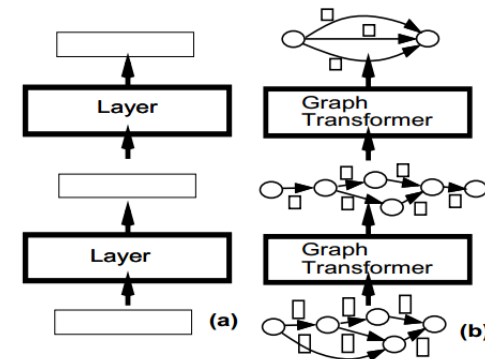
$$c(\mathbf{x}, \mathbf{y}) = \mathbf{w}^T \phi(\mathbf{x}, \mathbf{y}) = \sum_t \mathbf{w}^T \phi(\mathbf{x}, y_{t-1}, y_t)$$

- The likelihood decouples:

$$L(\mathbf{w}) = \sum_i \sum_t \mathbf{w}^T \phi(\mathbf{x}^i, y_{t-1}^i, y_t^i) - \log \sum_{\mathbf{y}} \exp \left(\sum_t \mathbf{w}^T \phi(\mathbf{x}^i, y_{t-1}, y_t) \right)$$

- Can use dynamic programming to compute costs (and derivatives)
- Extension to non-sequence models straightforward

Solution proposed in 1998



- Main insight: When layers are complex *graphs*, back-prop still works (LeCun et al 1998)
- This observation was recycled in 2001 under the name *Conditional Random Field*

Linear structured prediction

- Main idea: Define scores over cliques in a graph
- Example: sequence models may define a score

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- The likelihood decouples:

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Linear structured prediction

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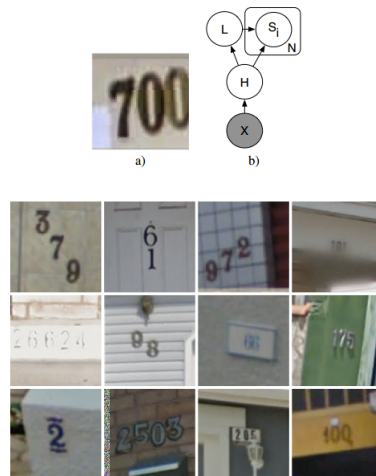
$$-\log \sum_{y_1 \dots y_{T-1}} \prod_{t=1}^{T-1} \sum_{y_T} \exp(\mathbf{w}^T \phi(\mathbf{x}^i, y_{T-1}, y_T))$$

- ▶ Can use dynamic programming to compute costs (and derivatives)
- ▶ Extension to non-sequence models straightforward

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Streetview (Goodfellow et al, ICLR 2014)



- ▶ see also: Jaderberg et al. ICLR 2015 for recognizing text in images

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Structured back-prop

- ▶ The reason for decomposition of cost/gradients is not linearity but additivity.
- ▶ We can use a neural network score

$$c(\mathbf{x}, \mathbf{y}) = \sum_t f_{\mathbf{w}}(\phi_t(\mathbf{x}, y_{t-1}, y_t))$$

- ▶ The log-likelihood then still decouples

$$L(\mathbf{w}) = \sum_i \sum_t f_{\mathbf{w}}(\phi(\mathbf{x}^i, y_{t-1}^i, y_t^i))$$

$$-\log \sum_{\mathbf{y}} \exp\left(\sum_t f_{\mathbf{w}}(\phi(\mathbf{x}^i, y_{t-1}, y_t))\right)$$

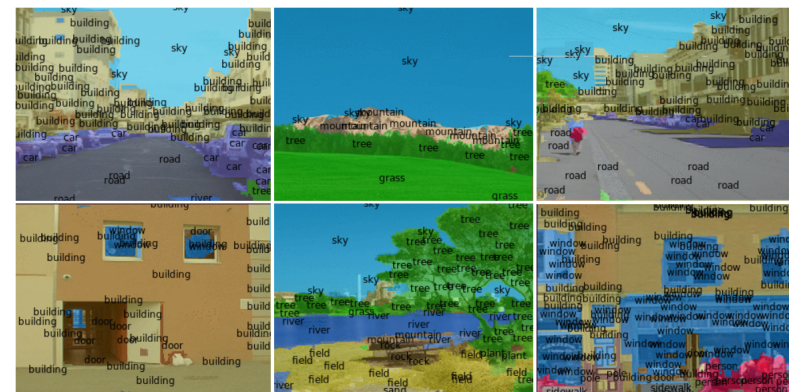
- ▶ ...and so does the gradient:

$$\frac{\partial L}{\partial \mathbf{w}} = \sum_i \sum_{\mathbf{y}} (\delta_{\mathbf{y}, \mathbf{y}^i} - p(\mathbf{y} | \mathbf{x}^i)) \sum_t \frac{\partial f_{\mathbf{w}}}{\partial \mathbf{w}}(\phi(\mathbf{x}^i, y_{t-1}^i, y_t^i))$$

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Towards scene understanding

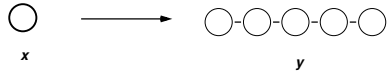


Farabet et al, 2013

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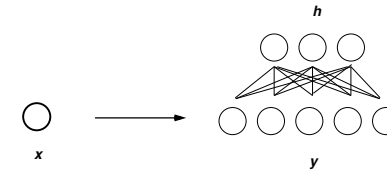
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A chance for unsupervised learning?



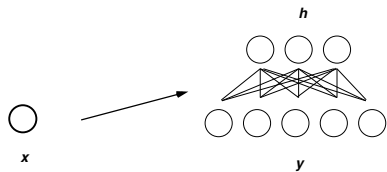
- ▶ A possible alternative to complicated graphical models on the outputs is to use *feature learning* on the outputs themselves.

A chance for unsupervised learning?



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A chance for unsupervised learning?



- ▶ A possible alternative to complicated graphical models on the outputs is to use *feature learning* on the outputs themselves.
- ▶ (...and potentially the dependence of the features themselves on the inputs, which would require 3-way connections)